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Psychometric and Machine Learning Approaches to Reduce the Length of Scales

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Abstract

Brief measures are important in psychology research because they reduce participant burden. Researchers can select items from longer measures either to build a short-form or to administer items conditional on a participant's previous responses. Researchers who carry out these item selection strategies either focus on estimating a precise score on the measure (typically carried out in a psychometric approach) or on predicting the score on the measure (possibly taking a machine learning approach). However, it is unclear how scores from the psychometric and machine learning approaches compare to each other. In this paper, the following four statistical approaches to select items are reviewed and illustrated: item response theory to build static short-forms, computerized adaptive testing, the genetic algorithm, and regression trees. Theoretical strengths and weaknesses between these four statistical approaches are discussed, and the overlap between the areas of psychometrics and machine learning is considered.

Keywords

Item response theory; machine learning; short-forms; tailored test

An important aim in the health and social sciences is to minimize respondent burden in questionnaire research (Gibbons et al., 2008). Previous research suggests that long surveys are associated with high nonresponse rates and poor response quality (Galesic & Bosnjak, 2009). Also, respondents may be more likely to participate and be more attentive during studies if scales are short rather than long (Credé, Harms, Niehorster, & Gaye-Valentine, 2012; Schroeders, Wilhelm, & Olaru, 2016). Human capital for social science studies can be expensive or time-limited, particularly in community or workplace samples, so investing time completing long measures can take away from the breadth of a study (Smith, McCarthy, & Anderson, 2000). Finally, administering fewer items can be convenient in longitudinal studies, large online studies, and personality studies in which items are either

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administered repeatedly or many constructs are measured at once (Sandy, Gosling, & Koelkebeck, 2014).

However, the potential advantages of giving respondents fewer items might not offset the disadvantages. Properties of the longer item set are unlikely to transfer to the reduced item set (Smith et al., 2000). A non-rigorous selection of items could lead to measures that misrepresent the original content and to scores with lower reliability, a different factor structure, or different correlations with external criteria compared to the scores from the full item set (Schroeders et al., 2016). Also, the breadth of assessment may depend on the consequences of test use (Messick, 1989). For example, an assessment for grade retention would be more comprehensive than a weekly assessment of reading ability. Finally, using scores from a reduced set of items in statistical models (e.g., regression) can affect overall conclusions. For example, if scores from the reduced set are used as control variables in regression, the control variable may account for less variance in the outcome and Type 1 errors for focal predictors may be encountered; if the score is used as a focal predictor, there might be less power to find significant effects and Type 2 errors may be encountered (Credé et al., 2012; Sahdra, Ciarrochi, Parker, & Scrucca, 2016).

Two common approaches to give fewer items to respondents are to administer either a short-form or a tailored test. In this paper, a static short-form refers to a fixed set of items from the longer set that is administered to all respondents; and a tailored test refers to a test in which respondents may receive different items depending on their previous responses. In psychology, researchers typically develop static short-forms and tailored tests using a psychometric framework with the ultimate goal of estimating precise scores on the whole item set. However, selecting items from a longer item set could be treated as a feature selection task, which is a common machine learning problem (Kuhn & Johnson, 2013). In theory, researchers could use machine learning algorithms to select the most important items that predict the score on the whole item set, and then respondents receive those items. In this paper, machine learning analogs of the psychometric approaches to select items are presented. The taxonomy in Figure 1 could provide a structure to discuss these statistical approaches.

The purpose of this paper is to review and provide a pedagogical illustration of psychometric and machine learning approaches to select fewer items and estimate a score on a unidimensional scale. In this paper, scales that estimate scores on reflective constructs are discussed, as opposed to scales primarily used for classification. Challenges to select items from scales that assess formative constructs are considered in the supplementary materials (Bollen, 2011; Murray & Booth, 2019). Although this paper focuses on the statistical procedures to select items, it is of utmost importance that readers consider many other methodological issues before launching into item selection. As such, this paper starts by describing some scale construction considerations. Then, psychometric approaches that use item response models to build static short-forms and adaptive tests are described. Next, machine learning approaches that build static short-forms using the genetic algorithm and tailored tests using regression trees are discussed. These four methods are then illustrated by selecting items from the Short Self-Regulation Questionnaire (Carey, Neal, & Collins, 2004), and the stability of the solution is studied with two small simulation studies (see supplementary materials). Finally, theoretical strengths and weaknesses of each approach are discussed.

Some Methodological Considerations before Selecting Items

Suppose that a researcher is developing a measure and has conceived the definition of the target construct and the structure of the test. Irwing and Hughes (2018) discuss several aspects that researchers should consider before choosing items to measure the construct: the number of items needed to measure the construct, the response scale, the scoring method for the item responses, the appropriate statistical model for the scale, the item piloting process, and the modality of administration. The last consideration is particularly important in this paper because tailored tests are typically administered via computer, and if computers are not available, then it might limit the methods that researchers could use to minimize respondent burden. Similarly, Smith et al. (2000) discuss other methodological considerations in the development of short-forms, such as validating the score interpretation from the full scale before dropping items, clarifying the goal of the short-form, and quantifying the resources and time saved by administering fewer items. Finally, in the content of short scales, Ziegler, Kemper, and Kruyen (2014) suggest that before using a scale, researchers should think clearly about the construct that the scale measures, the main purpose of the scale, and the target population for which the short scale is intended. For the purposes of illustrating the statistical procedures to select items, it is assumed that researchers have met the previous methodological considerations – mainly that the measure is relevant to the specific population to which it is administered and that we have access to a computer that can administer the tailored test.

Psychometric Approaches to Estimate Precise Assessment Scores

Psychometrics is the subfield of psychology concerned with using statistical models to measure psychological constructs and make inferences from psychological data (Jones & Thissen, 2006). Researchers who use psychometric approaches assume that a scale measures a latent construct. Important considerations when reducing the length of scales are that the items selected would yield scores that are reliable and that could be ascribed a valid interpretation. Reliability refers to the precision in measurement and is the proportion of observed score variance that is free of measurement error (Feldt & Brennan, 1989). Validity amounts to the pieces of evidence supporting the interpretation and use of a score (Messick, 1989), which includes examining content span, a dimensionality assessment of the items, and correlations of the measure with external criteria (Kane, 2006). A modern psychometric approach to build static short-forms or adaptive tests is based on item information functions estimated with item response models (Edelen & Reeve, 2007).

Building Static Short-forms using Item Information Functions.

Item response theory (IRT) refers to a family of statistical models used to analyze item responses and test scoring (de Ayala, 2009; Embretson & Reise, 2000). The parameters estimated in item response models are often used to study the psychometric properties of a measure and optimally select items to build a short-form (Edelen & Reeve, 2007). There are many different types of item response models, but this paper focuses on the graded response model (GRM; Samejima, 1969). The GRM is appropriate for Likert-type items commonly encountered in psychology. The GRM is formally expressed as (Edwards, Houts, & Li, 2018),

$$P\left(K_j = k_j \middle| \theta\right) = \frac{1}{1 + \exp\left[-a_j\left(\theta - b_{j(k)}\right)\right]} - \frac{1}{1 + \exp\left[-a_j\left(\theta - b_{j(k+1)}\right)\right]}. \tag{1}$$

The GRM produces trace lines that describe the probability of observing response k (i.e., a manifestation of K) on item j as a function of item parameters (a, b) and person parameters (θ). Here, θ represents the latent variable that the items assess. The a-parameter represents the strength of the relation between the item and θ, and the b-parameters represent the points along the θ range where the respondent has a 50% chance of endorsing item category k or a higher category. If an item consists of C item categories, then there are C - 1 b-parameters per item. By definition, the probability of endorsing the lowest category k or higher is 1, and the probability of endorsing category C + 1 is 0. If the item only has two categories, then the GRM reduces to the two-parameter logistic model (2PLM), another common IRT model not discussed here (see Embretson & Reise, 2000). IRT item parameters could be estimated using marginal maximum likelihood with the EM algorithm (Bock & Aitkin, 1981) or using Bayesian methods (Levy & Mislevy, 2016). Item parameters are calibrated on the same metric as the θ score, which provides a built-in mechanism to link scores regardless of which items are administered (Edelen & Reeve, 2007). Once item parameters are estimated, they could be used to calculate item information functions (IIF). Item information is related to measurement precision, or error in measurement, and indicates where each item discriminates the most across the range of θ. For example, some items may be more useful at distinguishing respondents low on θ as opposed to those high on θ. Items provide different amounts of information depending on the a- and b-parameters, where an item with a higher a-parameter provides more information than an item with a lower a-parameter at a fixed value of b. The IIF's are additive, so item information from all of the scale items can define a test information function (TIF), which in turn could be used to determine the θ range in which a set of items is most precise at measuring respondents. Using IIF's, researchers could build short-forms by selecting items depending on how much and where along the θ range each item is providing information.

Suppose that a researcher is interested in maximizing score precision for children in the clinical range of childhood aggression (e.g., one standard deviation above the θ mean of aggression), and score precision in the nonclinical range is not of interest. The general approach to build a static short-form using IRT follows four steps. First, the researcher conceptualizes a *target* TIF that peaks around one standard deviation above the θ mean. Then, items with IIF's that show high information one standard deviation above the θ mean are selected. Next, an empirical TIF from the selected items is estimated and compared to the *target* TIF (as a reference, see the top-left panel of the figure from the empirical example). As previously mentioned, the TIF is the sum of the item information functions. Finally, the researcher would evaluate if the empirical TIF sufficiently matches the target TIF, or if more items are needed to modify the shape of the empirical TIF (Edelen & Reeve, 2007; Hambleton, Swaminathan, & Rogers, 1991; Quinn et al., 2014). At this stage, researchers also could opt to manually include items that are considered important to the definition of the construct (Edelen & Reeve, 2007). The TIF from the static short-form would reflect less information (i.e., less measurement precision) than the TIF from the original item set because the static short-form includes fewer items. Then, the performance of the short-form could be validated in an independent sample by comparing the precision of the θ scores from the full form and the short-form, and the relation of the θ scores with external criteria. Using this general approach, researchers could build short-forms that match their specifications, such as a target length, specific content, measurement precision (i.e., information) across a specific θ range, and/or discrimination around a cut score.

Computerized Adaptive Testing.

The purpose of a computerized adaptive test (CAT) is to administer a test that is efficient, accurate, and *tailored* to each respondent (Embretson & Reise, 2000; Magis, Yan, & von Davier, 2017; van der Linden & Glas, 2006). For example, respondents who are low on θ are not likely to endorse any items that discriminate those who are high on θ, and respondents high on θ are likely to endorse all of the items that discriminate those who are low on θ. Therefore, it would not be efficient to administer all items if researchers know a priori which items the respondents are likely to endorse. A CAT uses an algorithm to select items conditional on a participant's previous responses, so the test adapts to the respondent and administers, in theory, only relevant items. By administering items adaptively, a CAT could cut administration time by at least 50% and still keep comparable score precision as if respondents took the whole item set (Van der Linden & Glas, 2006).

A CAT's general procedure is presented in Figure 2. First, an initial guess of the respondent's level of θ is needed. Lacking prior information, a reasonable guess is that the respondent is located at the mean of θ (Magis et al., 2017). Then, a CAT algorithm, which typically relies on item information functions, is used to decide which item to administer next. A response is then observed and θ is re-estimated. Item administration continues until the CAT meets a prespecified stopping rule. Before implementing a CAT, users need an item bank, which is a collection of items and item parameters that meets the assumptions of an item response model (for details on developing an item bank, see Revicki, Chen, & Tucker, 2013). For the current example, the item bank consists of the items from the preexisting unidimensional scale. Finally, users need three additional specifications to administer a CAT: the scoring method to estimate θ, the rule to select items, and the stopping rule.

A common method to estimate θ scores using IRT is *expected a posteriori* (EAP) scoring (Thissen & Steinberg, 2009). EAP[θ] scores are defined as the mean of a joint likelihood function L which is the product of two components: the product of the response probabilities (estimated from the item response model) and the assumed distribution of the latent variable ϕ(θ) (van der Linden & Pashley, 2010). Typically, the latent variable is assumed to be normally distributed, but it does not have to be (Woods, 2006). Formally, the joint likelihood L can be expressed as,

$$L = \prod_{j=1}^{J_{items}} p_j(K_j = k_j|\theta)\phi(\theta)\,. \tag{2}$$

Previous research suggests that EAP[θ] scores are more robust to IRT model misspecifications than other scoring methods and could prevent poor score estimation when the likelihood is bimodal. (Thissen & Wainer, 2001).

When the adaptive test starts, the first item administered is typically the item that is most informative around the mean of θ (Magis et al., 2017). After observing the response, then the next item is selected. There are many item selection strategies, but a common approach in large operational testing is to maximize Fisher information due to its simplicity and computational ease (Magis et al., 2017). If $\hat{\theta}_{t-1}(X_{t-1})$ is the θ estimate after observing response pattern X up to time t, then item j could be selected to maximize item information I at the current θ estimate,

$$j_t^* = \arg\max\ I_j\left(\hat{\theta}_{t-1}, (X_{t-1})\right). \quad (3)$$

There are other item selection rules that focus on either maximizing a different information function, maximizing a weighted information function, or using fully Bayesian methods. Each of these item selection rules have advantages and disadvantages, and are further discussed by Magis et al. (2017), van der Linden and Glas (2006), and van der Linden and Pashley (2010).

In theory, CAT could continue to administer items until it ran out of items, but this would defeat the purpose of CAT. A common stopping rule is based on the standard error of measurement (SEM) of the estimated EAP[θ] score (Walter, 2010). For example, if the stopping rule is a θ estimate with reliability of .90, one can administer items until the SEM ≤ .32 (note that reliability is 1 - SEM^2). This stopping rule yields θ estimates with comparable reliability. A time limit or a fixed number of items could also be used as stopping rules if score precision has not been achieved. For example, consider a situation in which a respondent has an extreme θ value. Item administration could stop early if the remaining items do not materially improve the score precision (van der Linden & Pashley, 2010). In this case, we might be mainly interested in determining if the respondent is very high or very low on θ.

Overall, the psychometric approaches to reduce the length of scales focus on score precision and use item information to select items to either build a static short-form or an adaptive test. The utility of these approaches depends on meeting the assumptions of the IRT model, such as local independence and correct dimensionality. If the assumptions are met, θ scores are comparable across respondents even if they receive different items or fewer items than the original item set. The linking property of IRT could make psychometric approaches to reduce the length of scales extremely valuable.

Machine Learning Approaches to Predict Assessment Scores

Machine learning is a subfield of artificial intelligence concerned with finding patterns in large datasets using data-driven algorithms and building models that are generalizable (Domingos, 2012; Yarkoni & Westfall, 2017). Models are typically built in a training dataset in which researchers are allowed to explore and determine relations, and then the model is cross-validated in testing data (e.g., data that was not used to estimate the model). Prediction performance in testing data provides an unbiased estimate of how well conclusions found in the training dataset are likely to generalize. Machine learning algorithms have been previously used to select which predictors to include in models (Kuhn & Johnson, 2013). Reducing the number of items from a preexisting scale can be treated as a feature selection problem in which items are chosen to predict respondent performance in the original item set. This procedure assumes that performance on the original item set could be summarized into a single score. Given the prevalence of use, computation ease, and a high correlation with IRT θ scores, a candidate statistic to summarize participant performance could be the summed score on the whole item set. In this case, the summed score is a composite fully determined by a linear combination of the item responses with unit weights. From a psychometric perspective, summed scores make strict assumptions about how items are related to the measured construct. However, machine learning approaches do not have to adhere to meeting psychometric assumptions (e.g., local independence) or to collapse response options that are rarely endorsed. For this application, the goal of machine learning is to make a short-form or a tailored test by selecting items that predict the summed score.

Building Static Short-forms using the Genetic Algorithm.

Suppose that a researcher wants to find a shorter item set that predicts participant performance in the original item set. If there are 30 items in the original item set, there are 2^{30} - 2 (i.e., more than a billion) candidate short-forms in the solution space. Evaluating all possible combinations of short-forms might not be a feasible task. One could turn to search algorithms to render a guided process to explore the candidate solutions (van der Linden, 1998). In machine learning, these types of problems are known as combinatorial optimization problems, and the genetic algorithm is one of many search algorithms that can be used to solve them (Marcoulides & Ing, 2012; van der Linden, 1998; Yarkoni, 2010). Previous research suggests that researchers have been successful at using the genetic algorithm to build static short-forms (Eisenbarth, Lilienfeld, & Yarkoni, 2014; Olaru, Witthoft, & Willhelm, 2015, Sahdra et al., 2016; Schroeders et al., 2016; Yarkoni, 2010).

The genetic algorithm starts by considering k random candidate short-forms to solve the optimization problem, and then uses q prespecified iterative steps to improve on those candidate short-forms by evaluating a user-defined criterion function. At the end of the iterative process, the genetic algorithm would provide the best answer (which might not be the right answer) out of those that it explored. The process by which the genetic algorithm moves through the solution space, tries out, and improves on the candidate short-forms mimics evolutionary principles found in nature (Goldberg, 1989; Holland, 1975). In essence, these evolutionary principles translate to functions that modify candidate short-forms (as in adding, dropping, or keeping items in the short-form) as the algorithm progresses through the solution space.

To make the example more manageable, suppose that researchers want to find an optimal short-form from a set of ten items (the procedure easily extends to more items). For convenience, candidate short-forms could be described using the following indicator form,

$$\begin{aligned}&1^{\text{st}}\ \text{short-form} = 1010100001 \text{ — items } 1, 3, 5, 7, \text{and } 10 \text{ are in the short-forms}\\&2^{\text{nd}}\ \text{short-forms} = 0010000100 \text{ — items 3 and 8 are in the short-form}\\&3^{\text{th}}\ \text{short-form} = 1101011101 \text{ — items } 1, 2, 4, 6, 7, 8, \text{ and } 10\\&\ \text{are in the short-form}\end{aligned} \tag{4}$$

In genetic algorithm terms, the indicator representation for each item (1 or 0) is referred to as a *gene*, and the indicator representation for the whole short-form is referred to as a *chromosome*. The genetic algorithm uses genetic operators, which are functions used on the chromosomes to modify the short-form as discussed above. The goal of genetic operators is to prevent the algorithm from getting struck on a seemingly good solution. An optimal short-form could be defined as one that maximizes the variance explained of the summed score of the ten items with the fewest number of items (Yarkoni, 2010). These criteria could be translated into a fit function that balances the residual variance of the summed score (i.e., 1 – adjusted R^2) and the length of the scale, such as,

$$Cost = l * s + \left(1 - R_{adj}^2\right). \tag{5}$$

In this case, s is the number of items in the scale, and l is the item cost, which penalizes the length of the measure to keep it as short as possible. Researchers could estimate the variance explained by predicting the summed score from individual items using OLS regression. Consequently, each chromosome is translated into an OLS regression model that has as many predictor items as there are genes with a value of 1. The genetic algorithm will explore the solution space to find a candidate short-form that minimizes the fit value in Eq. 5.

The general process to carry out the genetic algorithm is presented in Figure 3. The genetic algorithm starts with an initial random population of chromosomes (referred to as the *first generation*) and the fit value of each chromosome is estimated (using Eq. 5). Then, genetic operators are used to explore the solution space. The search process consists of four steps (Reeves, 2010): *selection, reproduction, evaluation*, and *replacement*. In the *selection* step, two chromosomes, referred to as parents, are chosen either randomly or by giving a higher probability of selection to chromosomes with a small fit value. In the *reproduction* step, two new offspring chromosomes are produced from the parent chromosomes. The reproduction step could take place using two commonly used genetic operators: the *crossover* operator and the *mutation* operator. The crossover operator swaps genes from the two parent chromosomes at a randomly chosen point,

$$\begin{aligned}&1^{\text{st}}\ \text{parent} = \mathbf{1001001000};\ \text{offspring} = \mathbf{100100}\mathit{1101}\\&2^{\text{nd}}\ \text{parent} = \mathit{1110101101};\ \text{offspring} = \mathit{111010}\mathbf{1000}.\end{aligned} \tag{6}$$

In this case, the crossover point is after the sixth gene. The last four genes of the 1^{st} parent are replaced by the last four genes of the 2^{nd} parent, and *vice versa*. Moreover, the mutation operator, with a prespecified probability, could randomly change each gene in the offspring chromosome to its complement, such as,

$$1^{\text{st}}\ \text{offspring} = 1\boldsymbol{0}01001000 \text{ to case} = 1\boldsymbol{1}01001000. \tag{7}$$

In this case, the second gene of the first offspring chromosome changed from a 0 to a 1, thus introducing diversity to the population of chromosomes. Crossover and mutation operators allow the genetic algorithm to strike a balance between the exploration of entirely new regions of the solution space and exploitation of visiting neighboring solutions (Črepinšek, Liu, & Mernik, 2013). The *reproduction* step continues until there are as many offspring chromosomes as in the initial population. Next, during the *evaluation* step, the fit value of each offspring chromosome is evaluated (again, using Eq. 5) and compared to the fit values of the chromosomes in the previous generation. Finally, in the *replacement* step, the set of chromosomes with the smallest fit values across both parent and offspring chromosomes move on to the next generation. The process continues for a prespecified *q* number of generations, and the best chromosome at the end of the iterative process determines the items that would compose the optimal static short-form.

Tailored Tests with Regression Trees.

Regression trees are popular machine learning models used for prediction (Breiman, Friedman, Olshen, & Stone, 1984; Hothorn, Hornik, & Zeileis, 2006; Strobl, Malley & Tutz, 2009). Regression trees provide a series of if-then rules that split the predictor space into regions in which the values on the outcome are most similar to each other. These regions are defined by predictor values and summarized using a decision tree (for reference, see the bottom-right panel of the figure from the empirical example). The predicted value in each of the predictor-defined regions is the mean value of the outcome. Previous research has used regression trees as an alternative to CAT when data are unlikely to meet IRT model assumptions (Magis et al., 2017, pg. 50; Yan, Lewis, & Stocking, 2004). Specifically, one could grow a regression tree on a training dataset to predict the summed score of the original item set from all individual item responses (Gibbons, Weiss, Frank, & Kupfer, 2016; Lu & Pektova, 2014; McArdle, 2013; Yan et al., 2004). Then, future respondents would receive items depending on the hierarchy of splits in the decision tree, similar to a CAT. Although the goal of CAT and trees is to administer items conditional on previous responses, an important difference is the extent to which the test is tailored to each respondent. With trees, a single and predetermined tree structure indicates the items that respondents receive, while in CAT the test changes as each respondent receives items. Previous research suggests that researchers have been successful at using trees for tailored testing (Gibbons et al., 2016; Fokkema, Smits, Kelderman, Carlier, & van Hemert, 2013; McArdle, 2013, Petway, 2010).

Regression trees are typically grown using a *binary recursive partitioning* algorithm. In general, the algorithm tries to find the best *binary* split among all possible predictor values to *partition* the dataset into two groups (regions) where outcome values are most similar. Then, the procedure is carried out *recursively* in each derived region until the algorithm meets a stopping point (e.g., a maximum number of splits, a minimum number of cases in each region, or no prediction improvement). Traditionally, the best split to define regions would be the split that minimizes the residual sums of squares based on the predicted (e.g., mean) and the observed values of the outcome in each region (Breiman et al., 1984). However, this approach has been criticized because the split search is not guided by statistical criteria, and the algorithm is biased toward selecting predictors with many possible splits (Strobl et al., 2009). That is, the possible number of splits per predictor depends on the predictor type. For a continuous predictor, candidate splits are considered along all unique values of the predictor. There are *k*-1 splits when the continuous predictor has *k* unique values. For unordered, categorical predictors, candidate splits are defined in the *one* (*or some*)-*versus-rest* form for all combination of predictors. There are 2^{k-1}-1 candidate splits when a predictor has *k* categories. These two criticisms could be addressed by using conditional inference trees (Hothorn, Hornik, & Zeileis, 2006). Instead of selecting the predictor that minimizes the residual sums of squares, a conditional inference tree selects predictors based on the relation between the predictors and the outcome. First, a permutation test per predictor is conducted, and a *p*-value is estimated. If at least one predictor has a *p*-value below a predetermined criterion (e.g., $p < .05$, although the *p*-value threshold could be treated as a tuning parameter), then the predictor associated with the smallest *p*-value is selected to split the examined region. After that, a split point on the selected predictor is determined by using a similar test statistic to what was used in the previous permutation test. Finally, the procedure is carried out recursively in the defined regions.

Conditional inference trees incorporate statistical criteria to select predicators via permutation tests and overcome variable selection bias by comparing predictors using a common metric: a *p*-value. Note that the algorithms that grow regression trees are considered greedy because once the algorithm decides on a predictor to split on, that decision is not reconsidered in further regions (i.e., a tree does not consider if a different first split could have been better for prediction in the long run for a specific learning problem). As a result, trees could lead to simple and unstable models that are unlikely to generalize to other datasets. Typically, researchers would use ensemble methods (i.e., random forest or boosting) or regularization to improve prediction accuracy (James et al., 2013). However, these methods do not provide the tree-like structure that is needed to determine the order in which the items are administered.

Overall, the machine learning approaches to reduce the length of scales are data-driven and do not make the same assumptions about the items that the psychometric approaches make. As implemented in this paper, both approaches focus on predicting the summed score from individual items by minimizing fit functions based on variance explained. A caveat is that the summed score is perfectly determined by the items responses, so including all of the items in the model explains all of the variance of the summed score. Therefore, algorithms that only minimize prediction error are expected to include all of the items because the prediction of the summed score is perfect when all of the items are in the model.[1] In the current application, the implementation of the genetic algorithm and the structure of the regression trees suggest that they are likely to select fewer items than the full item set – the stagewise selection of items by regression trees and the minimum threshold of short-form improvement by the genetic algorithm are likely to lead to item selection. Finally, note that the approach of each algorithm is different. Regression trees use a greedy

[1]A machine learning model often used for variable selection is regularized regression via Lasso (James et al., 2013). In a common implementation of the Lasso, model complexity is determined by minimizing the prediction error in test data. Suppose that a researcher uses the Lasso to predict the summed score from items. With this specific loss function, the Lasso might prefer a full rather than a sparse model because a model with all of the items would always minimize prediction error in test data. In order to use the Lasso to select items for a short-form, the determination of model complexity has to be more comprehensive. See supplementary materials for further discussion.

algorithm that selects the best item at each stage and does not reconsider the answer. On the other hand, the genetic algorithm is a dynamic, combinatorial algorithm that evaluates multiple simultaneous solutions and reconsiders each of the solutions until the algorithm stops running.

In this paper, the following four statistical approaches to select items and reduce the length of a preexisting unidimensional scale are illustrated: item response theory to build static short-forms, computerized adaptive testing, the genetic algorithm, and regression trees. The procedures will be evaluated by (1) comparing the items selected by each approach, (2) the recovery of the estimated score on the original item set by the four approaches, and (3) the relation between estimated scores from the original and reduced item set with external criteria. Furthermore, results from two small Monte Carlo simulations that evaluate the stability of the items selected and the variability of prediction accuracy are presented in the supplementary materials. The goal of this paper is to illustrate different psychometric and machine learning procedures to select items from a longer set.

Method

Motivating Example

Suppose that a researcher wants to administer the 31-item Short Self-Regulation Questionnaire (SSRQ; Carey, Neal, & Collins, 2004), but cannot administer all of the items. The original goal of the SSRQ and of its full form, the 63-item SRQ (Brown, Miller, & Lawendowski, 1999), was to measure self-regulation in substance-abusing samples in clinical settings, even though the wording of the items describes general self-regulation abilities that are not specific to substance use (see Table 1 for item prompts). Brown et al. (1999) indicate that scores from the SRQ correlate positively with drinking and negatively with impulsivity and risk-taking across diverse samples, such as a substance-abusing sample, a community sample, and a college sample. As such, it is expected that the SSRQ could be used to assess self-regulation in nonclinical samples, just like the one from this illustration. In a study by Eisenberg and colleagues (2019), 522 participants responded to around 600 items from popular self-regulation measures, including the SSRQ, to better understand the dimensions of the self-regulation construct. Items representative of the dimensions would then be used in other studies, such as presenting items to participants while being scanned in an fMRI machine and as part of an intervention to reduce binge eating and tobacco consumption (Eisenberg et al., 2018). Factor-analyzing 600 items might pose estimation issues, so to simplify the problem, items could be selected from each measure and then the reduced set could be factor-analyzed.

The dataset consists of 522 Mturk participants (52% women, 83% white) who completed the SSRQ among other questionnaires (Eisenberg et al., 2018). The dataset is publicly-available (see Eisenberg et al., 2019), but a curated version of the dataset with the items analyzed is included in the supplementary materials. The SSRQ items are rated on a five-point scale (strongly disagree to strongly agree), and item responses are typically summed to estimate a total score. Negatively-worded items were recoded so that higher responses indicate more self-regulation. During data collection, one item was administered twice[2] and one item was not administered, so only 29 out of 31 SSRQ items were examined. For the analyses, the dataset was split into a training dataset (N = 400) and testing dataset (N = 122).

Preliminary analyses

Item elimination and response collapsing were carried out in the training dataset before selecting items. The motivation of these preliminary analyses was that, if psychometric and machine learning approaches started with a dataset in which the estimation of each approach is feasible, then the observed differences would be due to the differences in item selection strategies (see supplementary materials for results without item elimination or response collapsing). In the training dataset, the lowest item response (1 = strongly disagree) was endorsed fewer than five times in 12 out of the 20 items. Therefore, the lowest item response was collapsed with the second lowest item response (2 = disagree). Similarly, the lowest item response for those 12 items was collapsed in the testing dataset. Parallel analysis (from the psych R-package; Revelle, 2018) on the training dataset suggested that the SSRQ is largely unidimensional (1st eigenvalue = 13.334, 2nd = 1.766, 3rd = 1.497). However, a unidimensional item factor model (fitted using the lavaan R-package with diagonally-weighted least squares; Rosseel, 2012) did not fit the training dataset well, $\chi^2(377) = 2{,}335.111$, $p<.001$, CFI = .980, RMSEA = .114. Modification indices suggested that some items in the training dataset might exhibit local dependence due to conceptual similarity (e.g., SSRQ28: *I have a lot of willpower*, and SSRQ29: *I am able to resist temptations*) or wording similarity (e.g., SSRQ10: *I learn from my mistakes*, and SSRQ05: *I don't seem to learn from my mistakes*). The Jackknife Slope Index (Edwards et al., 2018) identified six item clusters that exhibited local dependence (see supplementary materials). An item from each cluster was selected for further analysis, and the rest were discarded. As a result, 20 SSRQ items were used in this illustration, which exhibited an improved fit of the unidimensional item factor model, $\chi^2(170) = 464.207$, $p<.001$, CFI = .992, RMSEA = .066. Finally, item responses were recoded so that the lowest category was zero.

Procedure

The four item selection approaches were conducted in the training dataset using five-fold cross-validation (i.e., estimation of item response model parameters for the psychometric methods, finding items that minimized the fit function used by the genetic algorithm, and growing a regression tree). The overlap between the individual items selected across models was examined. Then, score recovery of the EAP[θ] of the full set of items (for the psychometric models) and the summed score (for the machine learning models) were evaluated. There were two criteria used to evaluate score recovery. The first criterion was the mean (and the cross-validation standard error; CVs.e.) of the mean squared difference (MSD) across folds,

$$MSD = E\left[(\hat{y}_i - y_i)^2\right]$$

[2]Only the mean of the two item responses was available and not the item response of each item. Preliminary analyses suggest that at most 86% of respondents gave the same response to the repeated item.

where $\hat{y}_i$ is the estimated score from the reduced set of items and y_i is the estimated score on the full set of items. Note that MSD estimates across approaches are on different metrics. The second criterion was the mean (and the CVs.e.) of the Pearson's correlation r between the estimated scores from the full set of items and the estimated scores from the reduced set, across folds. It is expected that the psychometric methods would be better at recovering the EAP[θ], and the machine learning methods would be better at recovering the summed score. Finally, the recovery of the scores for the testing dataset is presented in the supplementary materials.

Model Specification.—Below are the specifications of each item selection approach. Sample code for each of these analyses is presented in the supplementary materials.

1. In the training dataset, the graded response model (GRM; Samejima, 1969) was estimated in 4 (out of 5) folds using marginal maximum likelihood with the EM algorithm in the mirt R-package (Chalmers, 2012). Item information functions were then estimated using the item parameters. Items for the static short-form were manually selected to maximize test information across the range of θ and to assure content span. Finally, EAP[θ] scores were estimated from the static short-form in the left-out fold and the testing dataset.

2. Similar to above, item parameters estimated across the folds in the training data were used for adaptive testing. In the left-out fold and the testing dataset, a simulated CAT was carried out in the catR R-package (Magis et al., 2017). The EAP[θ] score for the set of 20 SSRQ items was used as the score that the CAT should recover. The specifications for the CAT were to score responses using EAP[θ] scoring; to select items that maximize Fisher information at the current θ estimate; and to stop item administration contingent on score precision so that the estimated EAP[θ] score SEM is at or below .30. For reference, an SEM of .30 yields an approximate score reliability of .90.

3. The genetic algorithm was carried out in 4 (out of 5) folds of the training dataset using the GAabbreviate R-package (Sahdra et al., 2016). There were 300 generations, the size of the population was 10, and the solution was constrained to have a maximum of 10 items, and these are specified by the user.[3] The crossover rate was .80, the mutation probability was .10, and *elitism* was specified, so the best chromosome was always conserved from generation to generation.[4] The fit function, presented in Eq. 5, was used to balance the variance explained on the summed score of the 20 SSRQ items and the length of the scale. The item cost was specified to be .01. Variance explained was estimated using OLS regression. After the genetic algorithm selected the most important items, the summed score of the 20 SSRQ items was regressed on the items remaining. The regression model was then used on the left-out fold and the testing dataset to estimate predicted summed scores of the 20 SSRQ items.

[3]The maximum number of items, generations, and population size have to be large enough so that the genetic algorithm explores many parts of the solution space. As shown in the illustration, a solution was found before generation 50, but more generations were carried out because of the low computational burden for this analysis.
[4]Note that these values are the defaults for the GAabbreviate R-package.

4. A conditional inference tree was grown in 4 (out of 5) folds of the training dataset to predict the summed score of the 20 SSRQ items using the partykit R-package (Hothorn & Zeileis, 2015). It was specified that a split could be conducted if the derived regions had at last ten cases and using 10-fold cross-validation to determine the p-value threshold that minimized prediction error. The tree was then used on the left-out fold and the testing dataset to estimate predicted summed scores of the 20 SSRQ items.

Correlations with Criteria.—The similarity of the relations of the scores from the original item set and the score from the four approaches with external criteria were investigated in the training dataset (see supplementary materials for correlations with scores in the testing dataset). The six criteria included were: nervousness and hopelessness (from the K6; Kessler et al., 2002), frequency of alcohol drinking, cups of coffee per day, and scores on the Brief Self-control Scale (BSCS; Tangney, Baumeister, & Boone, 2004) and the Short Grit Scale (Grit-S; Duckworth & Quinn, 2009). The difference in dependent correlations was assessed with the test by Williams (1959), and estimating Cohen's q as an effect size for the difference between two correlations. If the test is not significant and Cohen's q is small ($q \leq .10$), there would be evidence that the relation between the score from the reduced item set and criteria is similar to the relation between the score from the original item set and criteria.

Results

Recovering the Score from the Original Item Set

Scores from the Short-Form Built using Item Information.—To make scores comparable with other methods, eight items from the training dataset were chosen to build the static short-form, presented in Table 1. The first six items were selected to maximize scale information across the range of θ. Two items were then added because *thinking before acting* and monitoring food/alcohol consumption are expected to be relevant to the definition of the self-regulation construct (see supplemental materials for results without these items). The TIF from the original item set is taller than the TIF from the short-form (see Figure 4A), but both have a similar shape and discriminate approximately in the same range of the latent variable. The TIFs also suggest that the items do not provide a lot of information for participants high on the latent variable, so there is less precision in measuring participants high on self-regulation. For the recovery of the EAP[θ] score, the five-fold cross-validation mean MSD was .061 (CVs.e. = .004) and the mean r was .965 (CVs.e. = .005).

Scores from Computerized Adaptive Testing.—The average number of items selected by the CAT was 9.21 items, so the number of items that respondents received decreased by about 50%. The exposure rates of the items are presented in Table 1, suggesting that there were five items administered to almost all of the participants (which were the items with the highest a-parameters). The relationship between θ score and items administered are presented in Figure 4B, suggesting that participants high on the latent variable received more items, and that some participants did not meet the stopping rule of an SEM below .30. This is not surprising because the TIF from the original item set indicates that there are not a lot of items providing information on the higher range of the latent variable. For the recovery of the EAP[θ] score, the five-fold cross-validation mean MSD was .045 (CVs.e. = .008) and the mean *r* was .974 (CVs.e. = .006).

Scores from the Genetic Algorithm.—Eight items were selected by the genetic algorithm to build a short-form, presented in Table 1. Also, Figure 4C shows the relationship between the length of the optimal measure as the number of generations increased, suggesting that the genetic algorithm found that an eight-item static short-form could be an optimal solution by the 50th generation, and then the solution did not improve in the following 250 generations. The regression equation to predict the summed score of the original item set by the eight items selected by the genetic algorithm (GA_{score}) is,

$$GA_{score} = 6.81 + 2.16ssrq_1 + 1.83ssrq_3 + 2.31ssrq_4 + 2.84ssrq_{10} + 2.54ssrq_{15} + 2.06ssrq_{16} + 1.99ssrq_{17} + 2.22ssrq_{26}$$

In the recovery of the summed score, the five-fold cross-validation mean MSD was 9.122 (CVs.e. = .985) and the mean *r* was .970 (CVs.e. = .004).

Scores from trees.—The regression tree to predict the summed score of the original item set from the items is presented in Figure 4D. The *p*-value threshold was .05. If the tree were to be used with new respondents, the first item administered would be item SSRQ16, *I have a hard time setting goals for myself*. If respondents endorsed a 3 or less, participants would receive item SSRQ11, else respondents would receive item SSRQ4. Respondents would continue to make their way down the branches of the tree until they get to one of the leaves. Depending on the branches, each new respondent would receive between three and ten items. The unique items used by the tree are shown in Table 1. In the recovery of the summed score, the five-fold cross-validation mean MSD was 35.337 (CVs.e. = 3.557) and the mean *r* was .884 (CVs.e. = .011).

Comparing Scores and Items across Approaches

On average, there were around 8 to 9 items administered in each procedure. Table 1 suggests that the four procedures selected different items in the training dataset. For example, the item SSRQ16, *I have a hard time setting goals for myself*, was chosen by the four procedures, but the item SSRQ13, *I am able to accomplish goals for myself*, was not chosen by the genetic algorithm. Both items have high discrimination parameters and correlate *r*=.778, so it is likely that item SSRQ13 was not chosen by GA because item SSRQ13 and item SSRQ16 might account for similar variance of the summed score. On the other hand, the item SSRQ1, *I don't notice the effects of my actions until it's too late*, was selected by GA and trees, but not by the IRT short-form approach and was administered by the CAT only 14.8% of the time. Item SSRQ1 had a discrimination parameter below the average of those in the original item set, but it could be accounting for unique variance in the summed score.

Correlation profiles with criteria.—The top half of Table 2 shows the correlations among the EAP[θ] score of the original item set, the summed score of the original item set, and the scores from the four item selection methods for the training dataset using the estimated values of the five-fold cross-validation. Table 2 suggests that trees had poor recovery of the score of the original item set compared to the other three approaches. This might not be surprising because there are only 24 different scores that any given respondent could have obtained using tailored testing with regression trees, while there were over 100 unique scores using the other three approaches. Also, the bottom half of Table 2 shows the correlations between the scores from the original item set, the scores from the four item selection approaches, and six external criteria for the training dataset. The correlation profiles between the four approaches and the scores from the original item set were similar in magnitude and rank order. After a Bonferroni correction, 5 out of 24 correlations significantly differed between the scores from the original item set and the scores from the four approaches, and two of those correlations had a $q \geq 0.10$. The correlation between the regression tree predicted summed score and the BSCS was significantly different from the correlation between the summed score of the original items set and the BSCS ($q = .205$). Similar findings were observed for relation between the tree scores and the Grit-S ($q = .120$). Therefore, it is unlikely that the summed scores from the original item set and the tree scores are interchangeable to predict the BSCS and the Grit-S.

Discussion

Short-forms and tailored tests are important in applied studies because they reduce participant burden. This paper discussed four statistical procedures to build static short-forms and tailored tests that focus on estimating precise scores or on predicting the overall score on reflective measures. The psychometric approaches used item information from item response models to select items for a static short-form or decide the order in which items are administered in a CAT. In contrast, machine learning used exploratory approaches to select items that predict the summed score of the original item set. In the supplementary materials, simulation results suggest that the psychometric approaches consistently selected the same items across bootstrapped datasets. On the other hand, machine learning approaches did not select the same items consistently across bootstrapped datasets of size $N = 1{,}000$, and their prediction accuracy varied as a function of the sample size of the training dataset. However, our conclusions are limited to the conditions studied (i.e., samples of 200 to 1,000 cases). Also, in the supplementary materials, the challenges of selecting items that assess formative constructs were discussed. Although the machine learning approaches were the only methods out of those presented in the paper that could be used to select items in formative measures, we do not recommend the practice because 1) the interpretation of formative constructs is sensitive to the causal indicators in the model and 2) the predicted score would be a slight deviation from the mean if the items are uncorrelated. If researchers wanted to use psychometric methods to examine formative measures, a MIMIC model (not discussed here) might be a viable option (Bollen, 2011).

One could argue that the four methods produced predicted scores that are highly correlated with the scores from the full item set, so it would not matter what approach researchers use. Perhaps readers should not judge these four approaches by the end results, but rather by the process that each approach took to select items (Edelen & Reeve, 2007). Psychometric approaches focused on estimating a latent variable, which provides a mechanism for inference and interpretation of the construct. With a latent variable, researchers could estimate precise scores that could be ascribed a valid interpretation. On the other hand, the machine learning approaches focused on predicting the summed score of the measure by individual items. The machine learning approaches assumed that the summed score reflected participant performance on the measure. Therefore, if the outcome were to change, then the model that predicts the outcome is also likely to change. Machine learning approaches would favor items that account for unique variance in the summed score, and psychometric approaches would favor items that help estimate the θ score more precisely. In most machine learning applications, it is difficult to ascribe a valid interpretation to the model. However, in this specific instance the machine learning approaches were presented with a preexisting set of items which, theoretically, would have been vetted by content experts to create the scale. Therefore, one could have limited validity evidence to interpret the scores solely based on scale content.

Furthermore, selecting items from a preexisting item set might not be the best setting in which researchers could take full advantage of machine learning methodology. Machine learning approaches may be most useful in scenarios in which there are complex relations between the items and the criterion (performance summary measure) or in high-dimensional problems. Similarly, the psychometric approaches impose a latent variable model on the items and then estimate a score. Consequently, the performance of the psychometric approaches would depend on how closely researchers meet the assumptions of the item response model, such as specifying the correct number of dimensions and local independence. It might be possible that machine learning approaches could provide an alternative approach to reduce the length of scales if the data violate model assumptions (Yan et al., 2004), but this must be further examined.

Limitations and Future Directions

This paper only focused on demonstrating the statistical procedures to select items, and we did not discuss in detail the validation process of the reduced item set. Validity evidence is needed to determine if the SSRQ short-forms or adaptive tests illustrated fit a researcher's specific purpose. It is expected that the Mturk sample could reflect the population in which the SSRQ was developed (inpatient, community, or college samples), but this needs to be further tested. The sample used also is prone to inconsistent responding, which could have introduced some response biases (Barends & de Vries, 2019). Furthermore, there were several decisions about analyzing the empirical example that could have influenced the results. The data processing, such as collapsing rarely-endorsed item responses and eliminating locally dependent items, might have favored the psychometric methods – machine learning methods would have used the preprocessed data. The separation of the dataset into a training and a testing dataset would have favored the machine learning approaches – psychometric methods would prefer using all of the post-processed data for more precise item parameter estimates. The motivation of the data preprocessing decisions is that, if the item selection approaches started with different datasets, then differences in items selected and correlations with external criteria would be confounded by the differences in the algorithms and differences in the datasets. Finally, the flexibility of the four approaches was not fully demonstrated. The IRT approach for static short-forms was described as a manual approach, but other methods of *optimal test assembly* could select items that meet certain TIF conditions automatically (Harel & Baron, 2019). CAT could have included content balancing and item exposure controls, but these concerns might not be as relevant to psychology or medicine applications (Choi & Swartz, 2009) as they are to area of education. The genetic algorithm also could have optimized additional criteria besides variance explained, such as score reliability (Marcoulides & Drezner, 2001). Lastly, other algorithms that yield tree-like structures could have been investigated (Gibbons et al., 2016; Loh, 2014).

Future directions are to conduct an empirical study for a thorough analysis of the optimization criteria employed by these methods. The two frameworks presented in this illustration used different loss functions to select items, so it might be difficult to compare them in a simulation (but see the supplementary materials for some preliminary work in this area). There might not be a data-generating model that is fair to both approaches and what the benchmark of comparison would be, so more theoretical work is needed to solve these complexities. Perhaps a situation in which psychometric and machine learning approaches could be more directly compared would be in situations in which a scale is used for classification, such as using the CES-D to screen respondents for depression (Andresen, Malmgren, Carter, & Patrick, 1994). In this case, a machine learning approach could predict the probability of a positive diagnosis directly from the items, while a psychometric approach would estimate a θ score and decide if it is above a cut score (Gonzalez, in press). Then, sensitivity and specificity to predict the diagnosis could be compared across the two frameworks. Also, it would be interesting to examine how these methods handle multidimensionality. There are analogs to item information in multidimensional item response models (Reckase, 2009) or one could use marginal trace lines (Stucky, Thissen, & Orlando Edelen, 2013). The genetic algorithm could use structural equation modeling and a multi-objective fit function to select a short-form per dimension (Marcoulides & Drezner, 2001). There are also tree analogs for multivariate outcomes (De'Ath, 2002, Hothorn & Zeileis, 2015).

An overarching goal of this paper was to illustrate areas in which psychometrics and machine learning intersect, such as in assessment development. Some problems faced by machine learning are similar to those in psychometrics, such as classifying individuals, balancing of exploratory and confirmatory analyses, dimension reduction, and variable selection. With this paper, researchers are encouraged to think about the intersection of psychometrics and machine learning for assessment and how these areas could learn from each other.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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	Static Short-form	Tailored Testing
Precision	Item Response Theory	Computerized Adaptive Testing
Prediction	Genetic Algorithm	Regression Trees

Figure 1.
Taxonomy of approaches to reduce the length of scales.

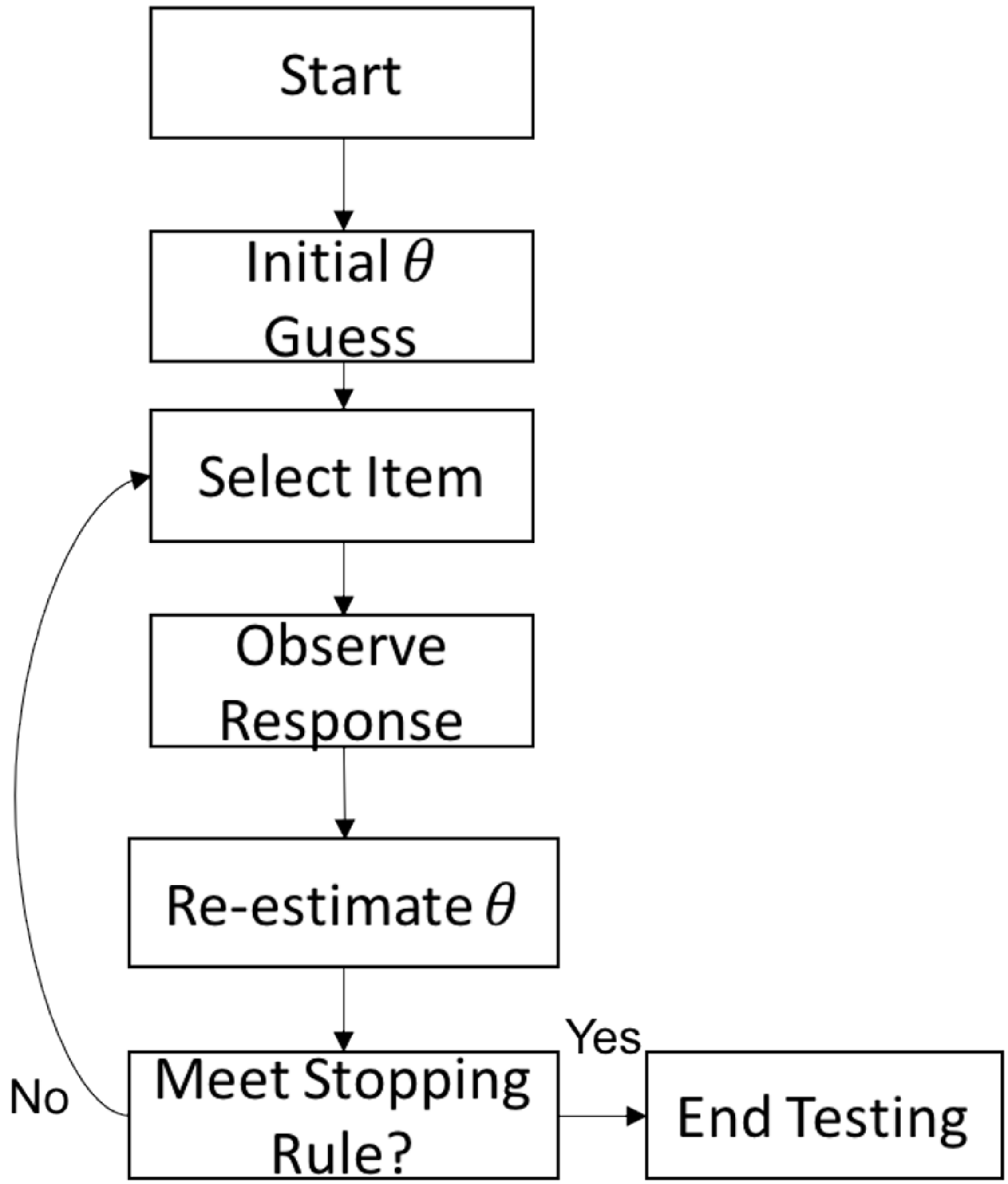

Figure 2.
General procedure for computerized adaptive testing.

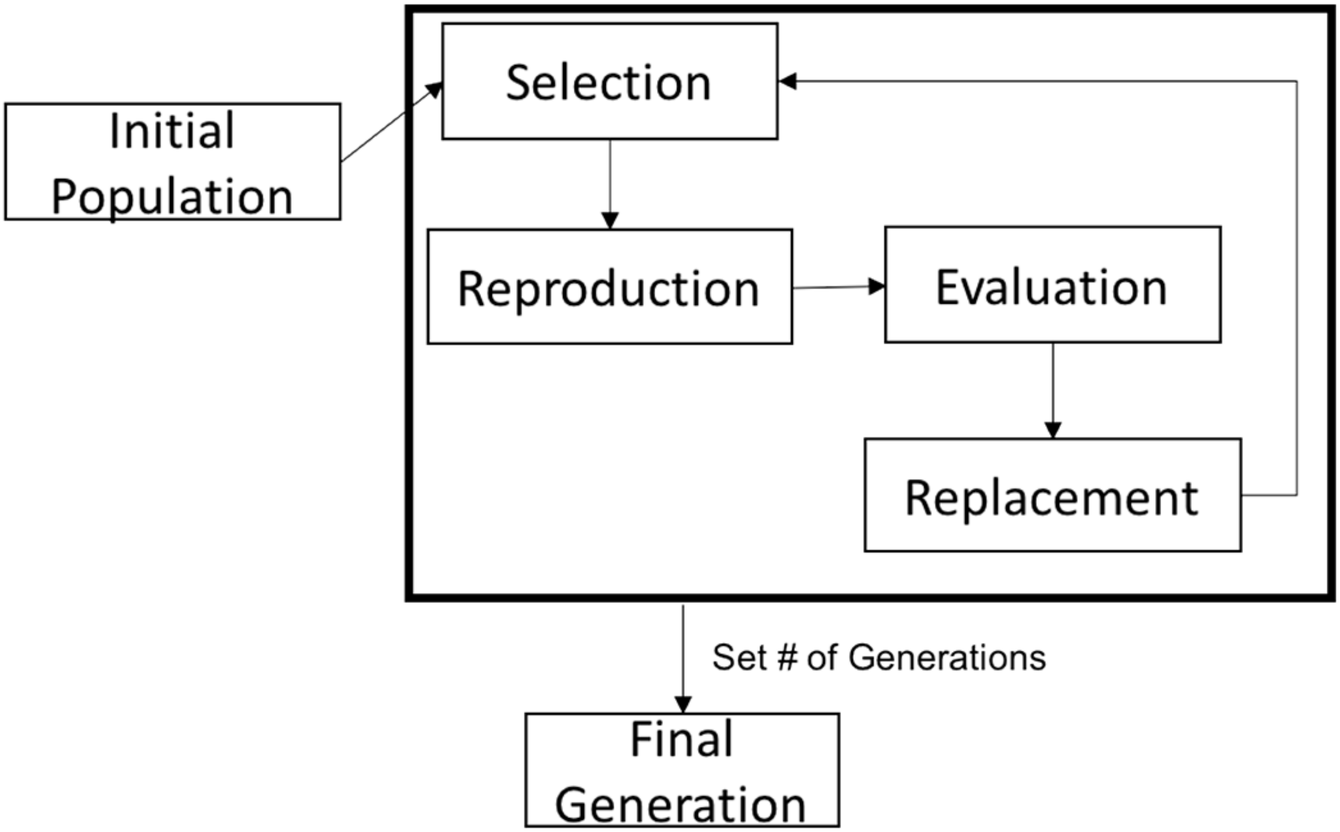

Figure 3.
General steps for the genetic algorithm.

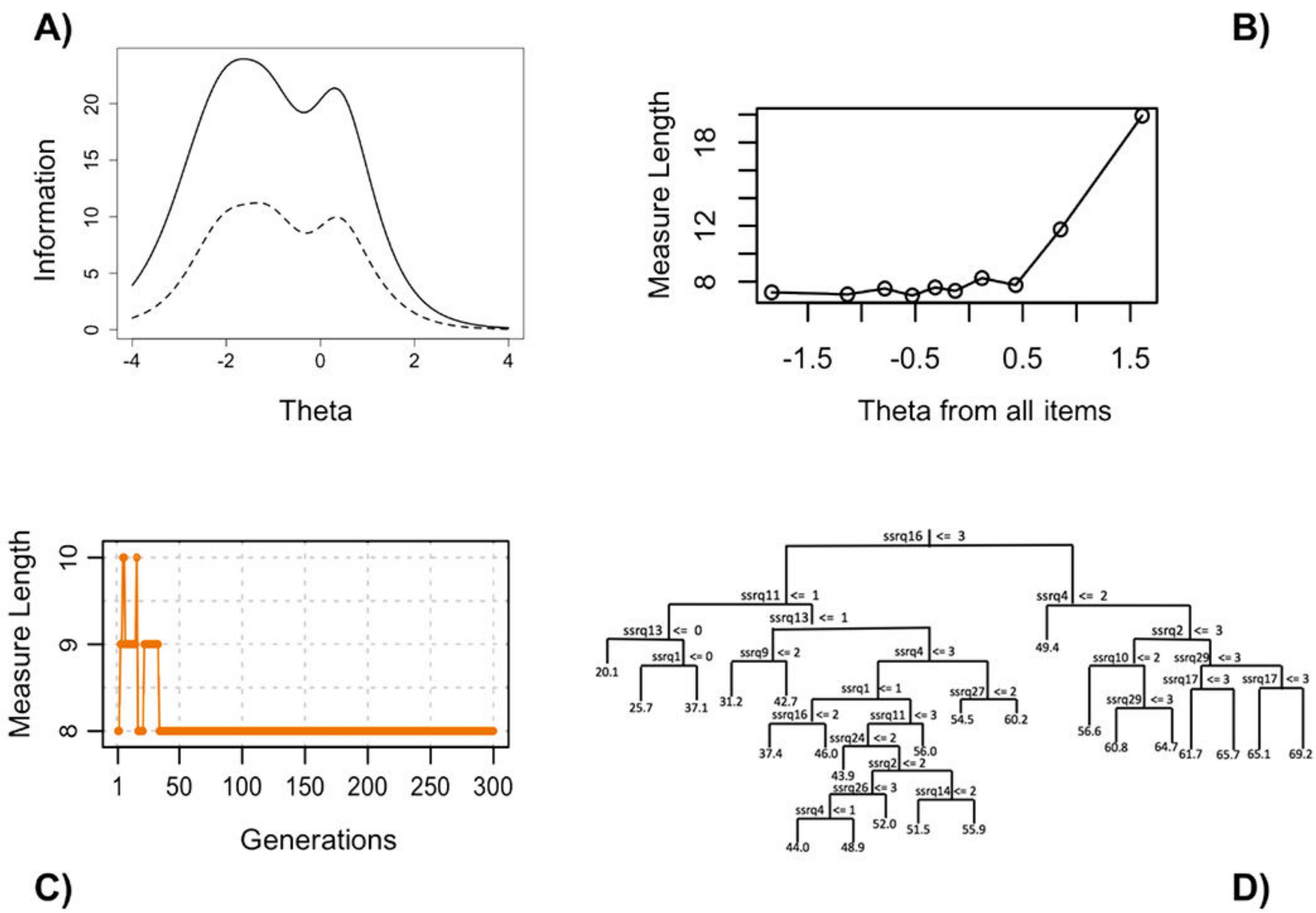

Figure 4.

Summary plot of the performance of the four machine learning algorithms. A) shows the test information functions for the original item set (solid, 20 items) and reduced item set (dashed, 8 items). B) shows the relationship between items administered and theta estimate. C) shows the relationship between the length of the measured as a function of the number of generations (iterations). D) shows the regression tree to predict the SSRQ summed score by individual SSRQ items. Note: SSRQ=short self-regulation questionnaire.

Table 1.

SSRQ items administered by each of the four item selection approaches.

Labels	Items	IRT SF items	CAT exposure rate	GA items	Tree nodes
SSRQ01	I don't notice the effects of my actions until it's too late.		0.148	X	X
SSRQ02	I put off making decisions.	X	0.541		X
SSRQ03	It's hard for me to notice when I've 'had enough' (alcohol, food, sweets).	Y	0.148	X	
SSRQ04	I have trouble following through with things once I've made up my mind to do something.		0.992	X	X
SSRQ07	I can usually find several different possibilities when I want to change something.		0.123		
SSRQ08	Often I don't notice what I'm doing until someone calls it to my attention.		0.148		
SSRQ09	I usually think before I act.	Y	0.098		X
SSRQ10	I learn from my mistakes.		0.746	X	X
SSRQ11	I give up quickly.	X	1.000		X
SSRQ12	I usually keep track of my progress toward my goals.		0.426		
SSRQ13	I am able to accomplish goals for myself.	X	1.000		X
SSRQ14	I have personal standards, and try to live up to them.		0.221		X
SSRQ15	As soon as I see a problem or challenge, I start looking for possible solutions.	X	0.459	X	
SSRQ16	I have a hard time setting goals for myself.	X	1.000	X	X
SSRQ17	When I'm trying to change something, I pay a lot of attention to how I'm doing.		0.189	X	X
SSRQ21	I know how I want to be.		0.148		
SSRQ24	I tend to keep doing the same thing, even when it doesn't work.		0.311		X
SSRQ26	If I wanted to change, I am confident that I could do it.		1.000	X	X
SSRQ27	I can stick to a plan that's working well.		0.328		X
SSRQ29	I am able to resist temptation.	X	0.189		X
	Average number of items administered	8 items	~9 items	8 items	3-10 items

Note: SSRQ=Short Self-Regulation Questionnaire; X=item chosen; Y=item included for content coverage; IRT SF=item response theory short-form; CAT=Computerized Adaptive Testing (maximize Fisher's information); GA=genetic algorithm. Not all SSRQ items are numbered sequentially because some items that exhibited local dependence were discarded (see supplementary materials).

Table 2.

Correlations between estimated scores of the SSRQ, summed scores of the SSRQ, and outcomes in the training dataset using 5-fold cross-validation.

	EAP[θ]	Summed Score	IRT SF EAP[θ]	CAT EAP[θ]	GA score	Tree score
EAP[θ]	1.000					
Summed Score	0.968	1.000				
IRT SF EAP[θ]	0.967	0.943	1.000			
CAT EAP[θ]	0.976	0.928	0.965	1.000		
GA score	0.937	0.970	0.916	0.900	1.000	
Tree score	0.881	0.881	0.884	0.879	0.865	1.000
Nervousness	−0.227	−0.257	−0.260	−0.236	−0.269	−0.259
Hopelessness	−0.300	−0.353	−0.339[a]	−0.310	−0.347	−0.327
Alcohol Frequency	−0.018	−0.020	−0.017	0.003	−0.022	0.010
Coffee per Day	0.073	0.068	0.071	0.078	0.060	0.054
BSCS	0.738	0.756	0.744	0.717	0.730[a]	0.654[a]
Grit-S	0.776	0.762	0.789	0.794	0.734[a]	0.707[a]

Note:

[a] is for correlations significantly different from the summed score after a Bonferroni correction (p=.002).

EAP[θ]=expected a posteriori θ estimate; IRT SF=item response theory short-form; CAT= Computerized Adaptive Testing (maximize Fisher's information); GA=genetic algorithm; BSCS=Brief Self-control Scale; Grit-S=Short Grit Scale